



Feature Tree:

- Chaitanya - Assignment 7.2 (Long and Thin) <<Default>_Display State 1>
 - Solid Bodies(1)
 - Chamfer1
 - Equations
 - "Shaft Length" = 70.00mm
 - "Brace Diameter" = 16.00mm
 - "Shaft Diameter" = 12.00mm
 - Material <not specified>
 - Front Plane
 - Top Plane
 - Right Plane
 - Origin
 - Revolve1
 - Sketch1
 - Cut-Revolve1
 - Sketch2
 - Cut-Extrude1
 - Sketch3
 - Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
 - 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
 - Chamfer1

Configurations

- ▼   Chaitanya - Assignment 7.2 Configuration(s) (Long and Thin)
 - ▶  Tables
 -   Long and Thin [Chaitanya - Assignment 7.2]
 -  — Short and Thick [Chaitanya - Assignment 7.2]



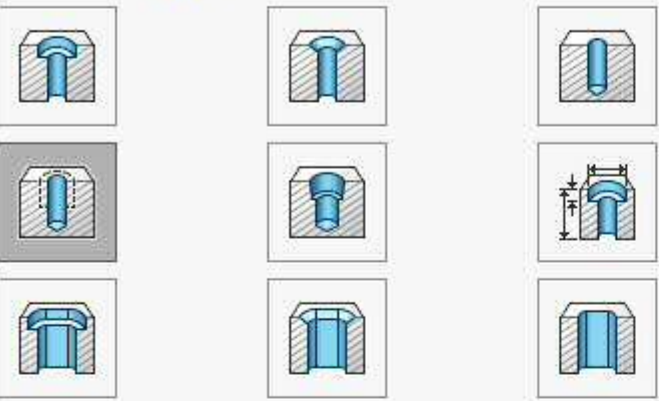
Threaded Hole Wizard:

 Type

 Positions

No Favorite Selected

Hole Type



Standard:

ANSI Inch

Type:

Tapped hole

Hole Specifications

Size:

3/8-24

☐ Show custom sizing

End Condition

 Through All

Thread:

Through All

Options

☒ With thread callout

☐ Thread class

☐ Near side countersink

☐ Far side countersink

Tolerance/Precision

Callout value

Thru Tap Drill Diameter (8.43)

+.01

1.50

-.01

None

.01

X.XXX

.01

.12 (Document)



Hole Specification



Through-All Hole Wizard:

Type



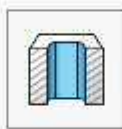
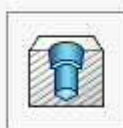
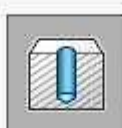
Positions

Favorite



No Favorite Selected

Hole Type



Standard:

ANSI Metric

Type:

Dowel Holes

Hole Specifications

Size:

Ø1.0



Show custom sizing



6.35mm

End Condition



Through All

Options

☐ Near side countersink

☐ Far side countersink

Tolerance/Precision

Callout value

Diameter (6.35)

+0.01
1.50
-0.01

None

.01
X.XXX
.01

.12 (Document)

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

☐ MMGS (millimeter, gram, second)

IPS (inch, pound, second)

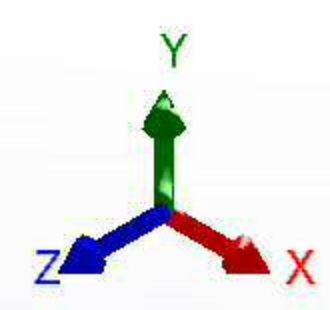
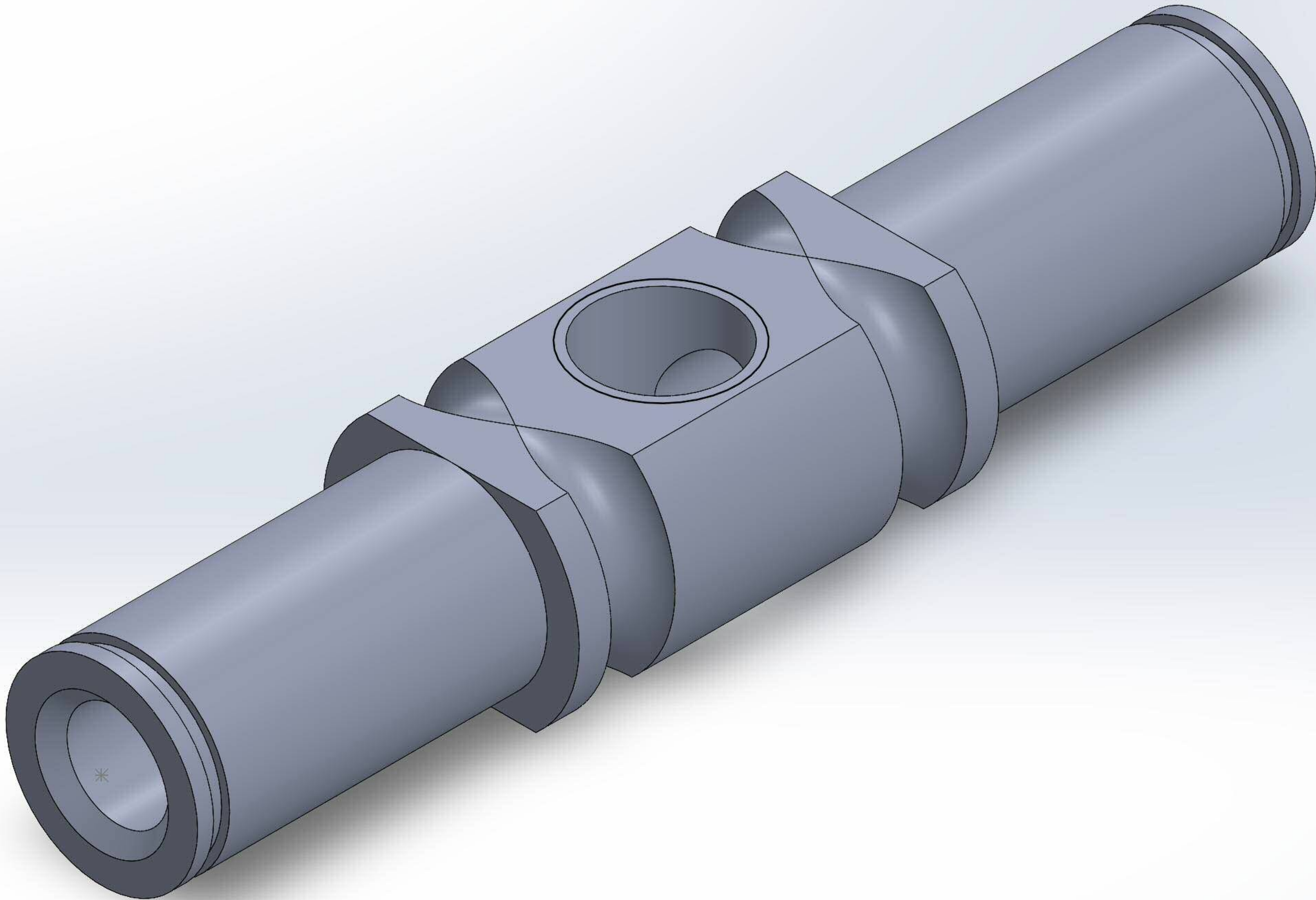
Edit Document Units...

Simplified Interface

MMGS



Isometric View (L&T):



Chaitanya - Assignment 7.2 (Long and Thin)

Equations

- "Shaft Length" = 70.00mm
- "Brace Diameter" = 16.00mm
- "Shaft Diameter" = 12.00mm

Material <not specified>

Front Plane

Top Plane

Right Plane

Origin

Revolve1

- Sketch1

Cut-Revolve1

- Sketch2

Cut-Extrude1

- Sketch3

Ø6.35mm Dowel Hole1

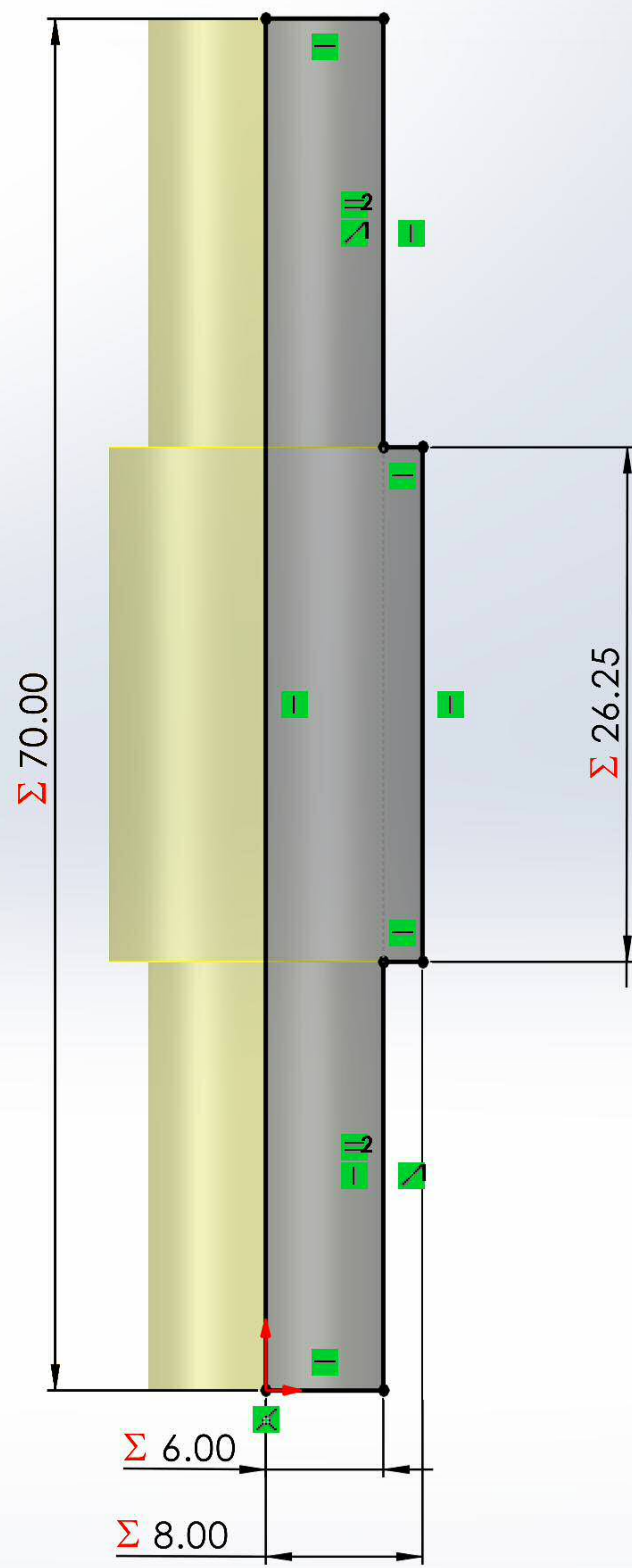
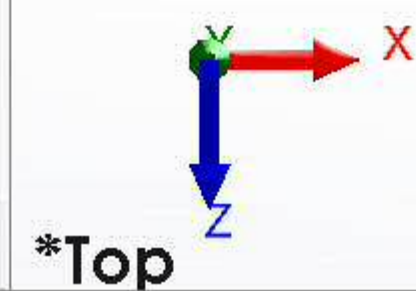
- Sketch4
- Sketch5

3/8-24 Tapped Hole1

- Sketch6
- Sketch7
- Hole Thread1

Chamfer1

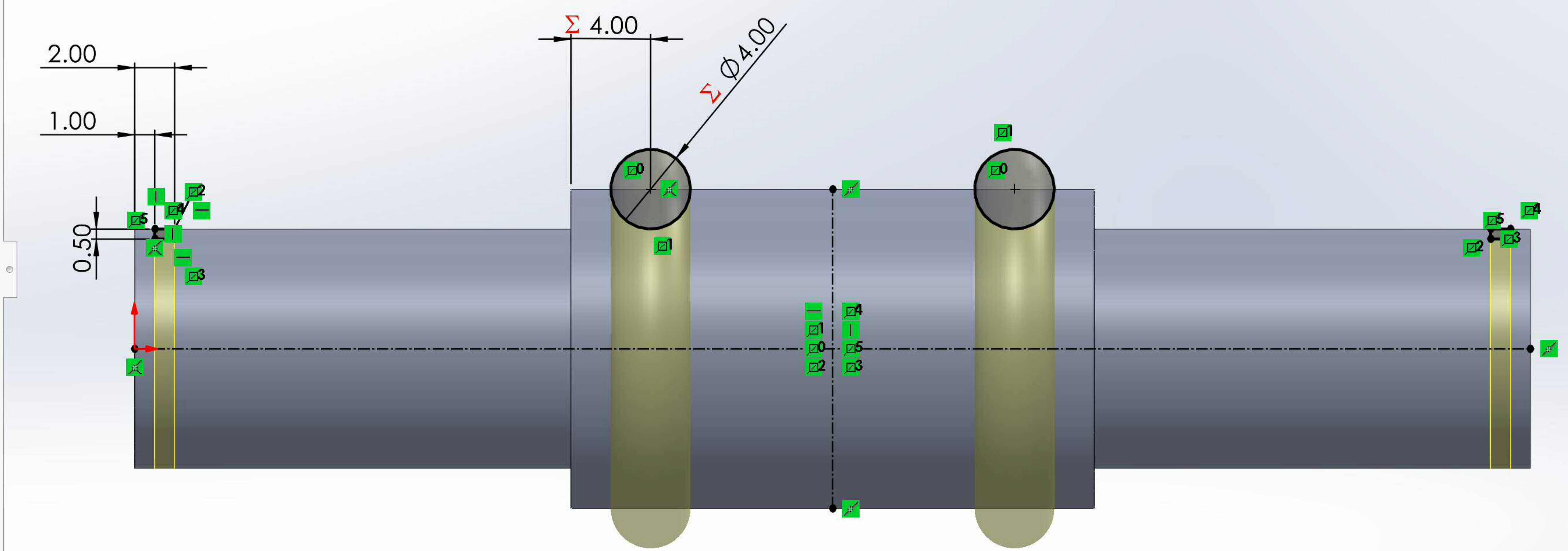
Sketch 1 (L&T):



Chaitanya - Assignment 7.2 (Long and Thin) <

- Solid Bodies(1)
 - Revolve1
- Equations
 - "Shaft Length" = 70.00mm
 - "Brace Diameter" = 16.00mm
 - "Shaft Diameter" = 12.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 2 (L&T):





Chaitanya - Assignment 7.2 (Long and Thin) <

Solid Bodies(1)

Cut-Revolve1

Equations

"Shaft Length" = 70.00mm

"Brace Diameter" = 16.00mm

"Shaft Diameter" = 12.00mm

Material <not specified>

Front Plane

Top Plane

Right Plane

Origin

Revolve1

Sketch1

Cut-Revolve1

Sketch2

Cut-Extrude1

Sketch3

Ø6.35mm Dowel Hole1

Sketch4

Sketch5

3/8-24 Tapped Hole1

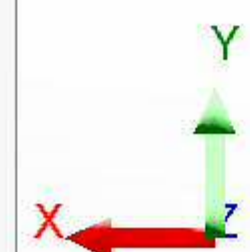
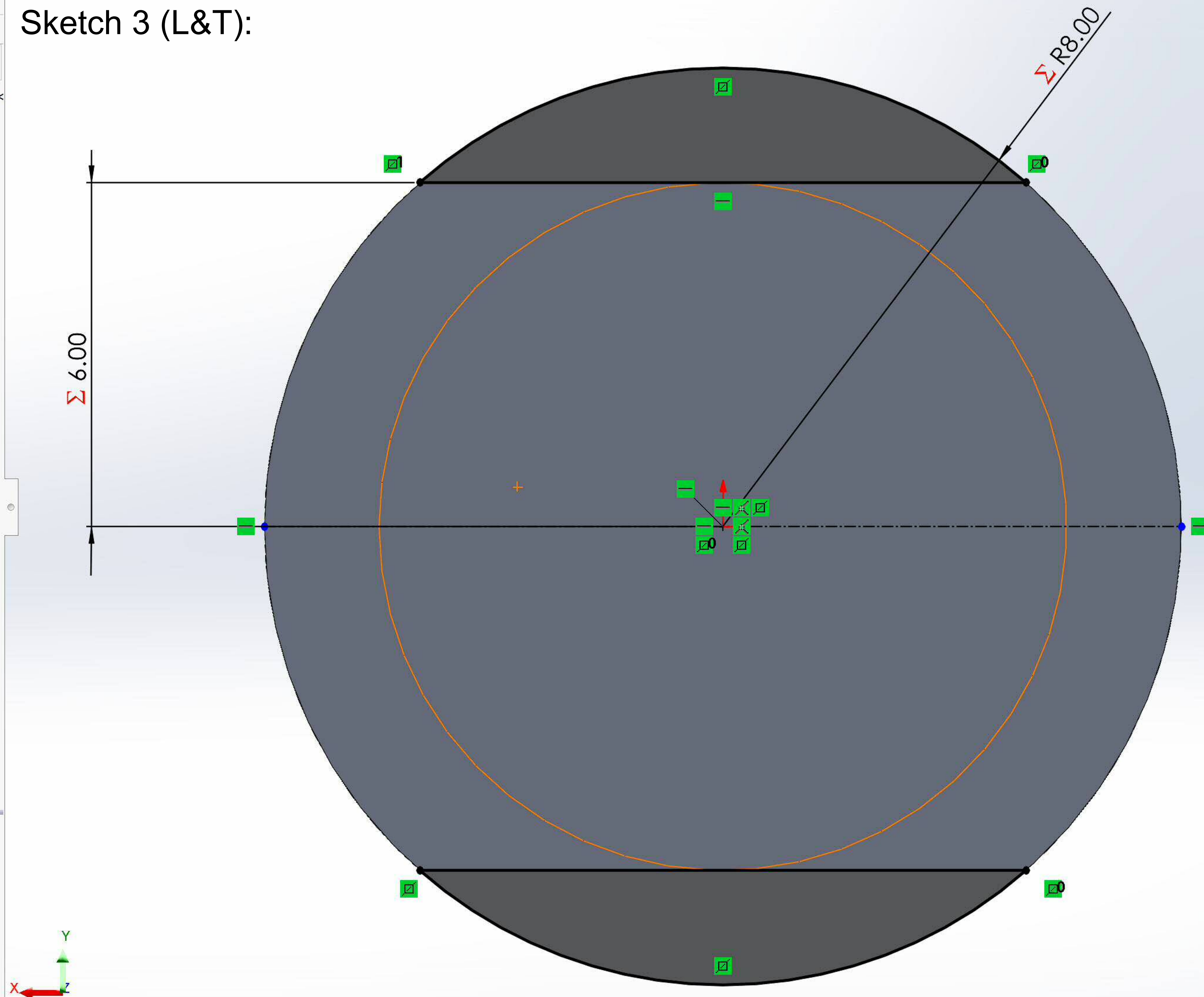
Sketch6

Sketch7

Hole Thread1

Chamfer1

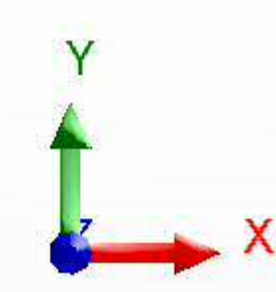
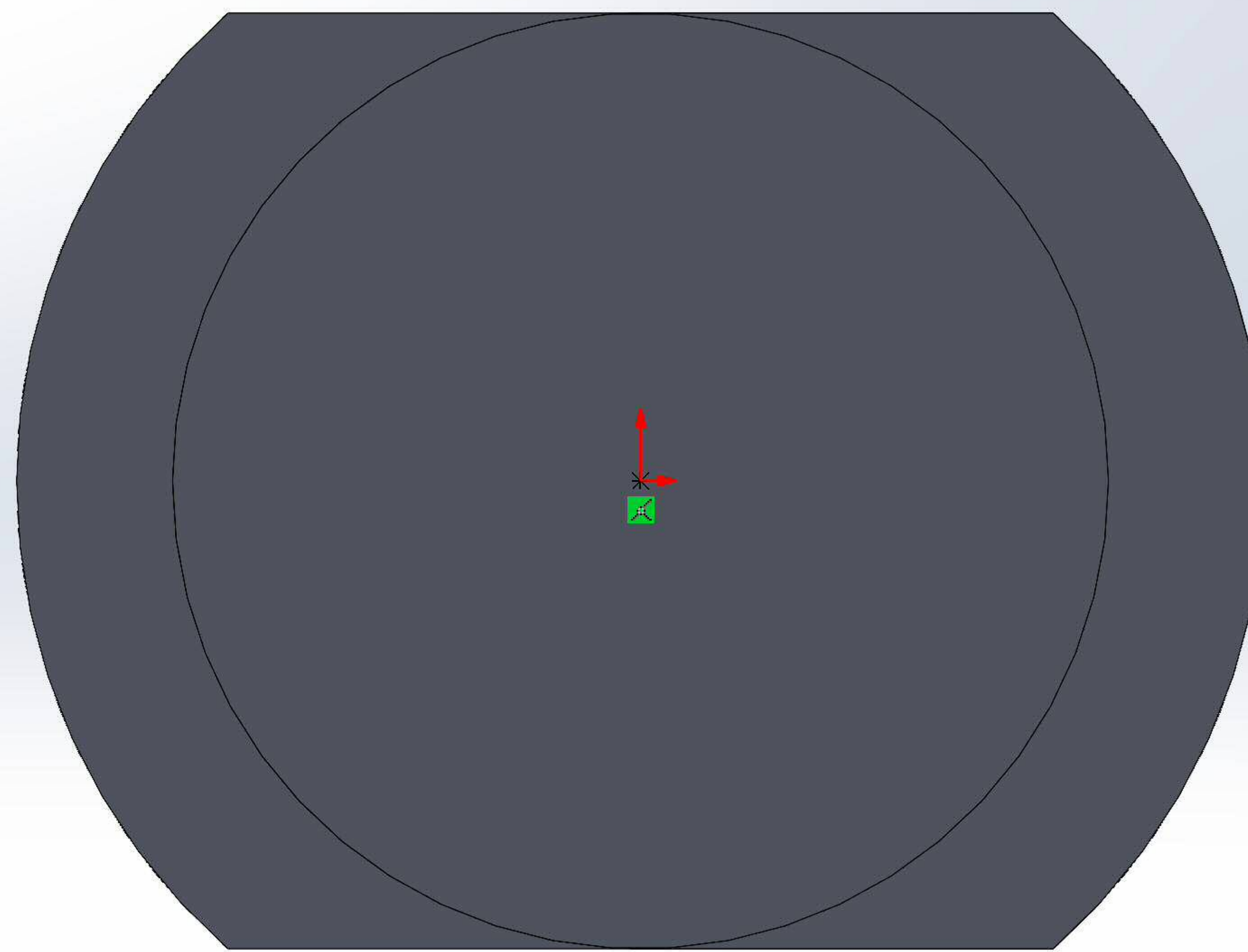
Sketch 3 (L&T):



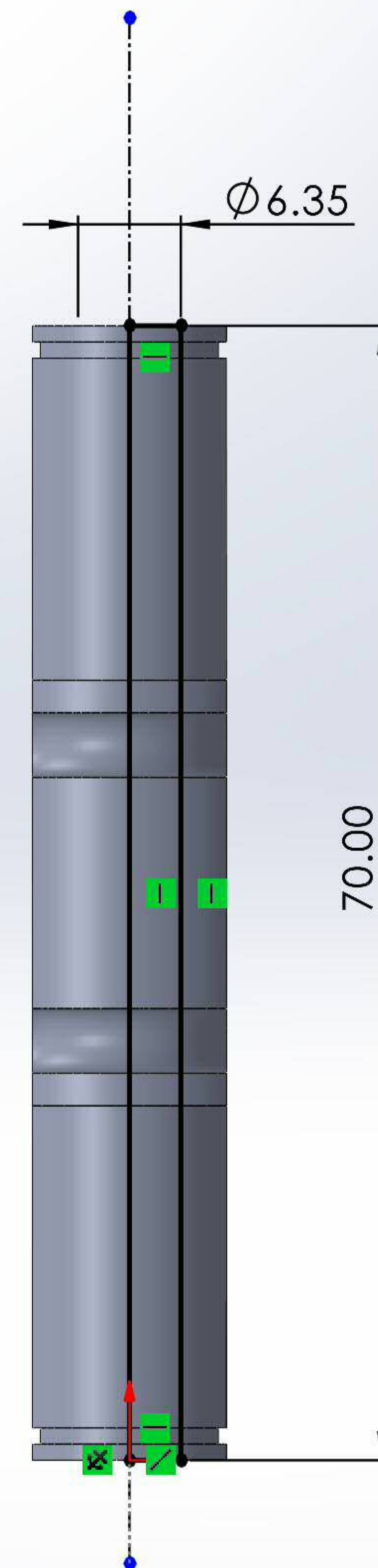
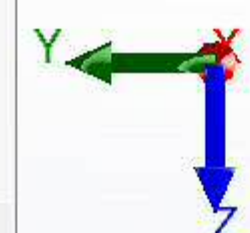
Feature Tree:

- Chaitanya - Assignment 7.2 (Long and Thin) <
- Solid Bodies(1)
 - Cut-Extrude1
- Equations
 - "Shaft Length" =70.00mm
 - "Brace Diameter" =16.00mm
 - "Shaft Diameter" =12.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
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 - Sketch5
- 3/8-24 Tapped Hole1
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 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 4 (L&T): Created By Hole Wizard



*Front



Chaitanya - Assignment 7.2 (Long and Thin) <

Solid Bodies(1)

- Ø6.35mm Dowel Hole1

Equations

- "Shaft Length" = 70.00mm
- "Brace Diameter" = 16.00mm
- "Shaft Diameter" = 12.00mm

Material <not specified>

Front Plane

Top Plane

Right Plane

Origin

Revolve1

- Sketch1

Cut-Revolve1

- Sketch2

Cut-Extrude1

- Sketch3

Ø6.35mm Dowel Hole1

- Sketch4
- Sketch5

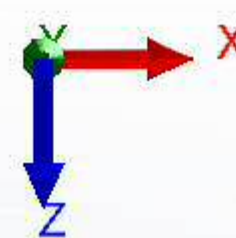
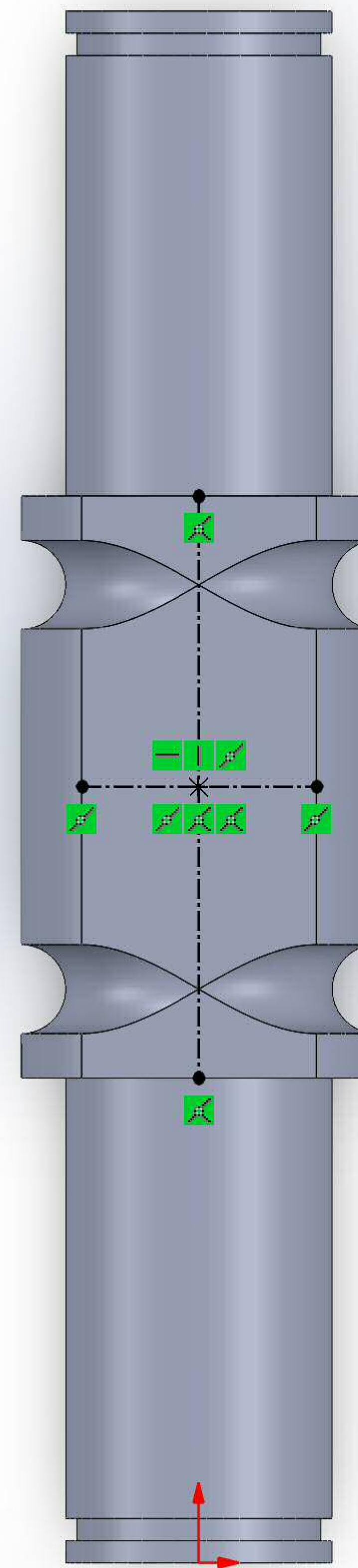
3/8-24 Tapped Hole1

- Sketch6
- Sketch7

Hole Thread1

Chamfer1

Sketch 6 (L&T): Created For Hole Wizard





Chaitanya - Assignment 7.2 (Long and Thin) <<

Solid Bodies(1)

Ø6.35mm Dowel Hole1

Equations

"Shaft Length" = 70.00mm

"Brace Diameter" = 16.00mm

"Shaft Diameter" = 12.00mm

Material <not specified>

Front Plane

Top Plane

Right Plane

Origin

Revolve1

Sketch1

Cut-Revolve1

Sketch2

Cut-Extrude1

Sketch3

Ø6.35mm Dowel Hole1

Sketch4

Sketch5

3/8-24 Tapped Hole1

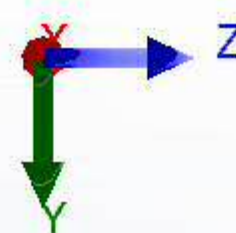
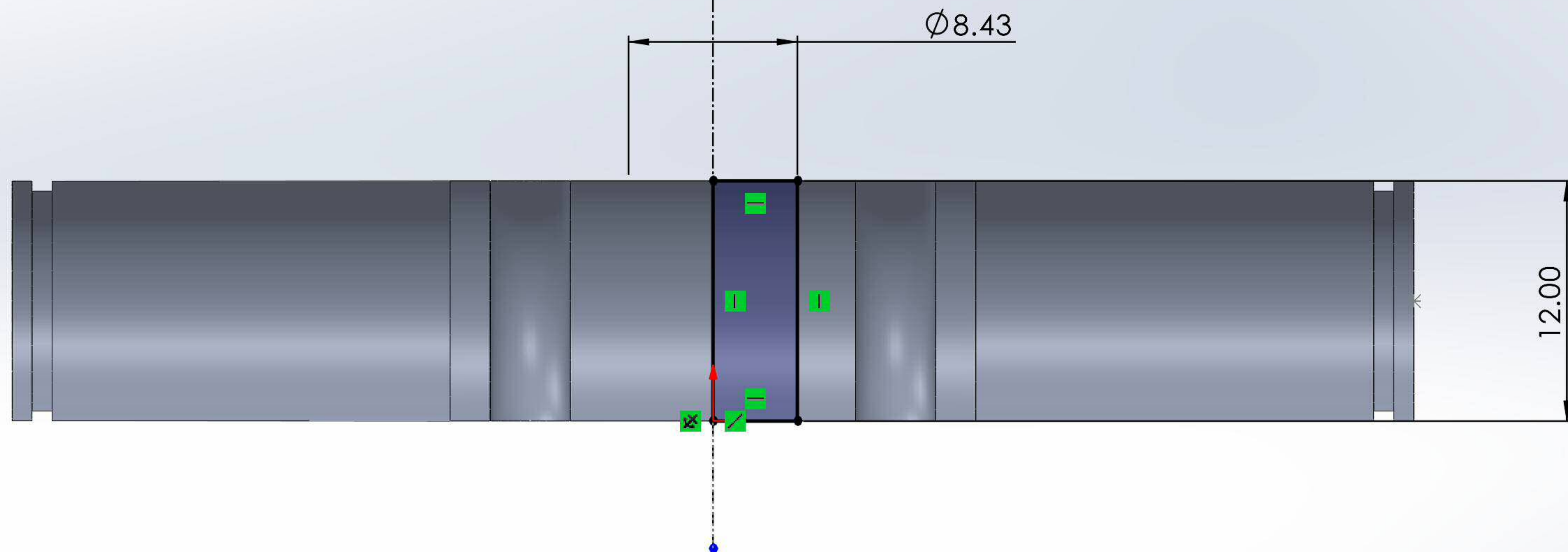
Sketch6

Sketch7

Hole Thread1

Chamfer1

Sketch 7 (L&T): Created By Hole Wizard



Global Variables & Equations (L&T):



Filter All Fields



Long and Thin



Name	Value / Equation	Evaluates to	Comments
Global Variables			
"Shaft Length"	= 70mm	70.00mm	
"Brace Diameter"	= 16mm	16.00mm	
"Shaft Diameter"	= 12mm	12.00mm	
Add global variable			
Features			
Add feature suppression			
Equations			
"D1@Sketch1"	= "Shaft Length"	70mm	
"D2@Sketch1"	= ("Shaft Length") * (3 / 8)	26.25mm	
"D3@Sketch1"	= "Shaft Diameter" / 2	6mm	
"D4@Sketch1"	= "Brace Diameter" / 2	8mm	
"D5@Sketch2"	= "Brace Diameter" - "Shaft Diameter"	4mm	
"D2@Sketch3"	= "Shaft Diameter" / 2	6mm	
"D1@Sketch3"	= "Brace Diameter" / 2	8mm	
"D4@Sketch2"	= "Brace Diameter" - "Shaft Diameter"	4mm	
Add equation			

OK

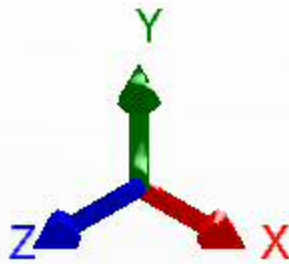
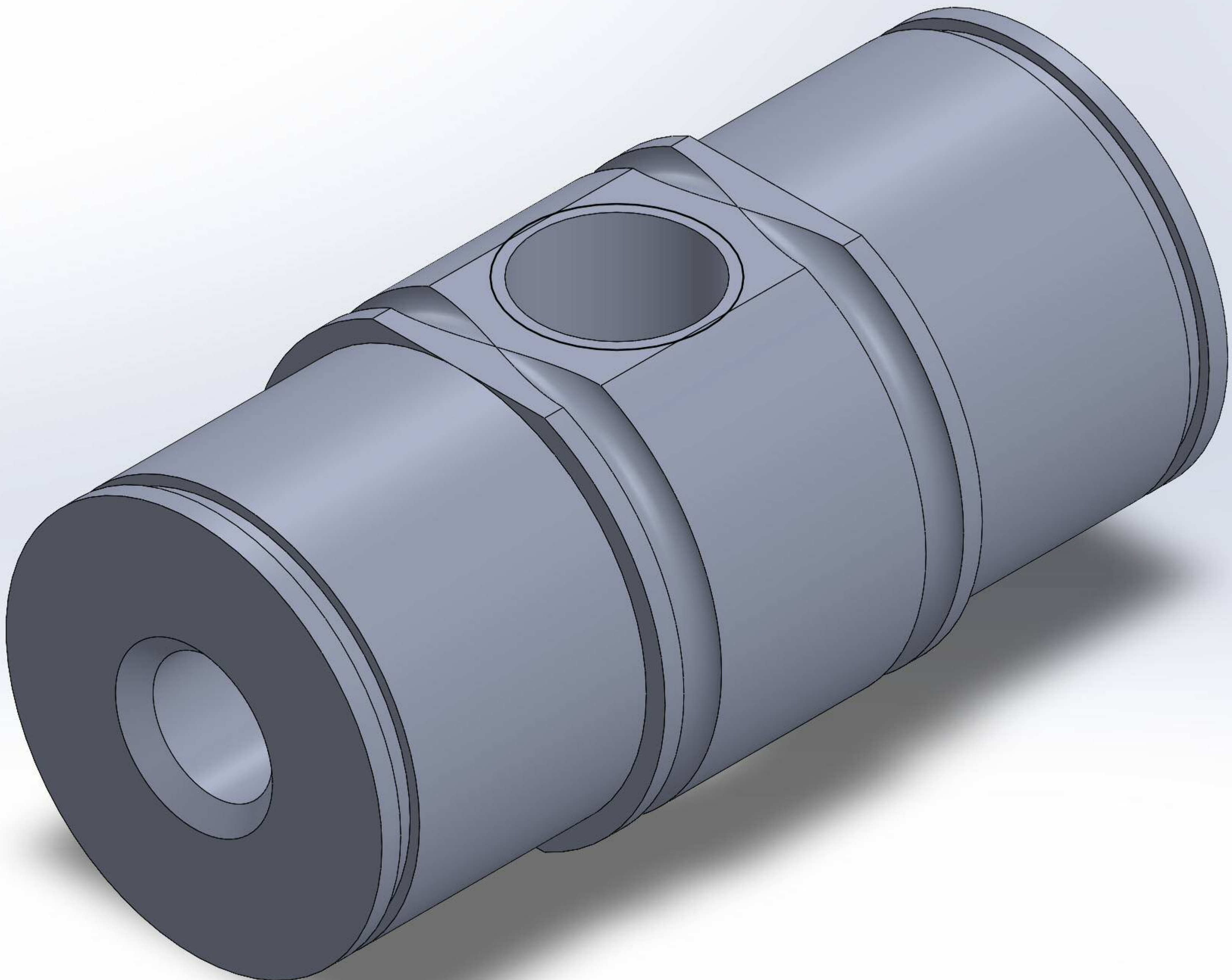
Cancel

Import...

Export...

Help

Isometric View (S&Th):



Chaitanya - Assignment 7.2 (Short and Thick)

Equations

- "Shaft Length" = 45.00mm
- "Brace Diameter" = 22.00mm
- "Shaft Diameter" = 20.00mm

Material <not specified>

Front Plane

Top Plane

Right Plane

Origin

Revolve1

Sketch1

Cut-Revolve1

Sketch2

Cut-Extrude1

Sketch3

Ø6.35mm Dowel Hole1

Sketch4

Sketch5

3/8-24 Tapped Hole1

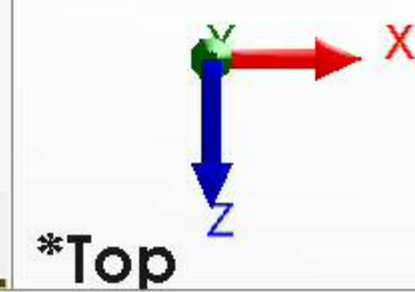
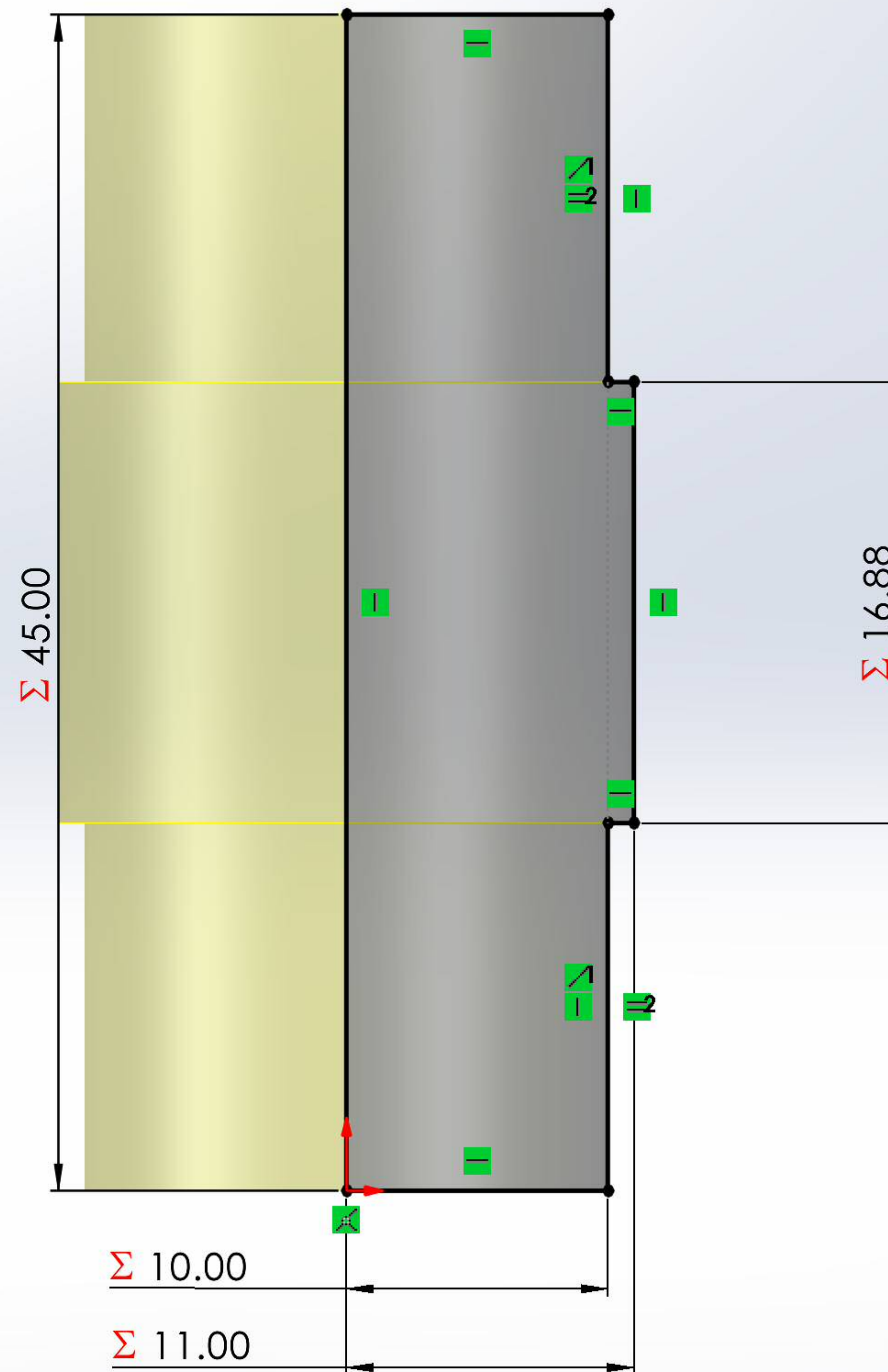
Sketch6

Sketch7

Hole Thread1

Chamfer1

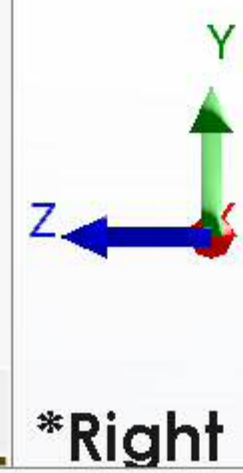
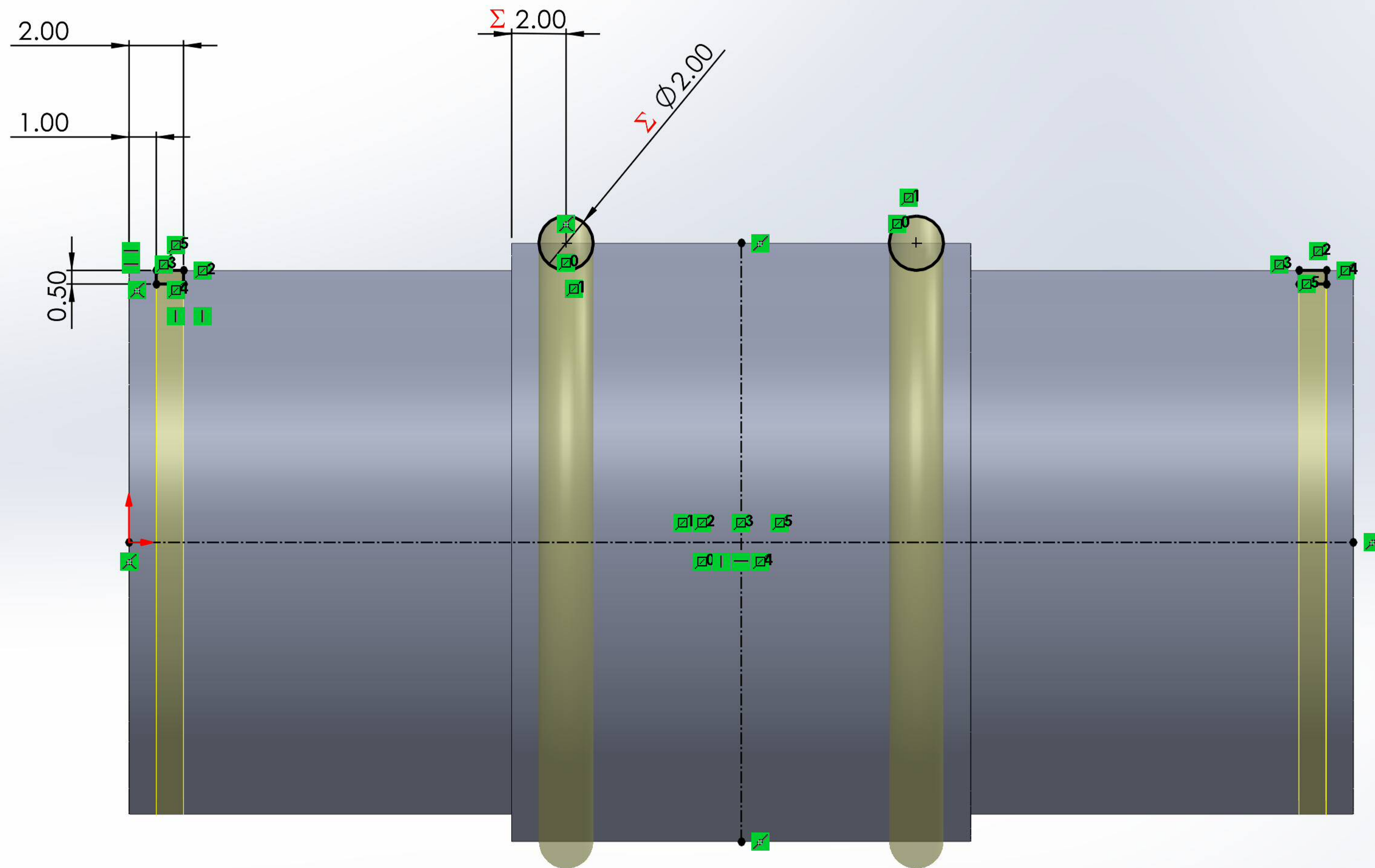
Sketch 1 (S&Th):



Chaitanya - Assignment 7.2 (Short and Thick)

- Solid Bodies(1)
 - Revolve1
- Equations
 - "Shaft Length" =45.00mm
 - "Brace Diameter" =22.00mm
 - "Shaft Diameter" =20.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

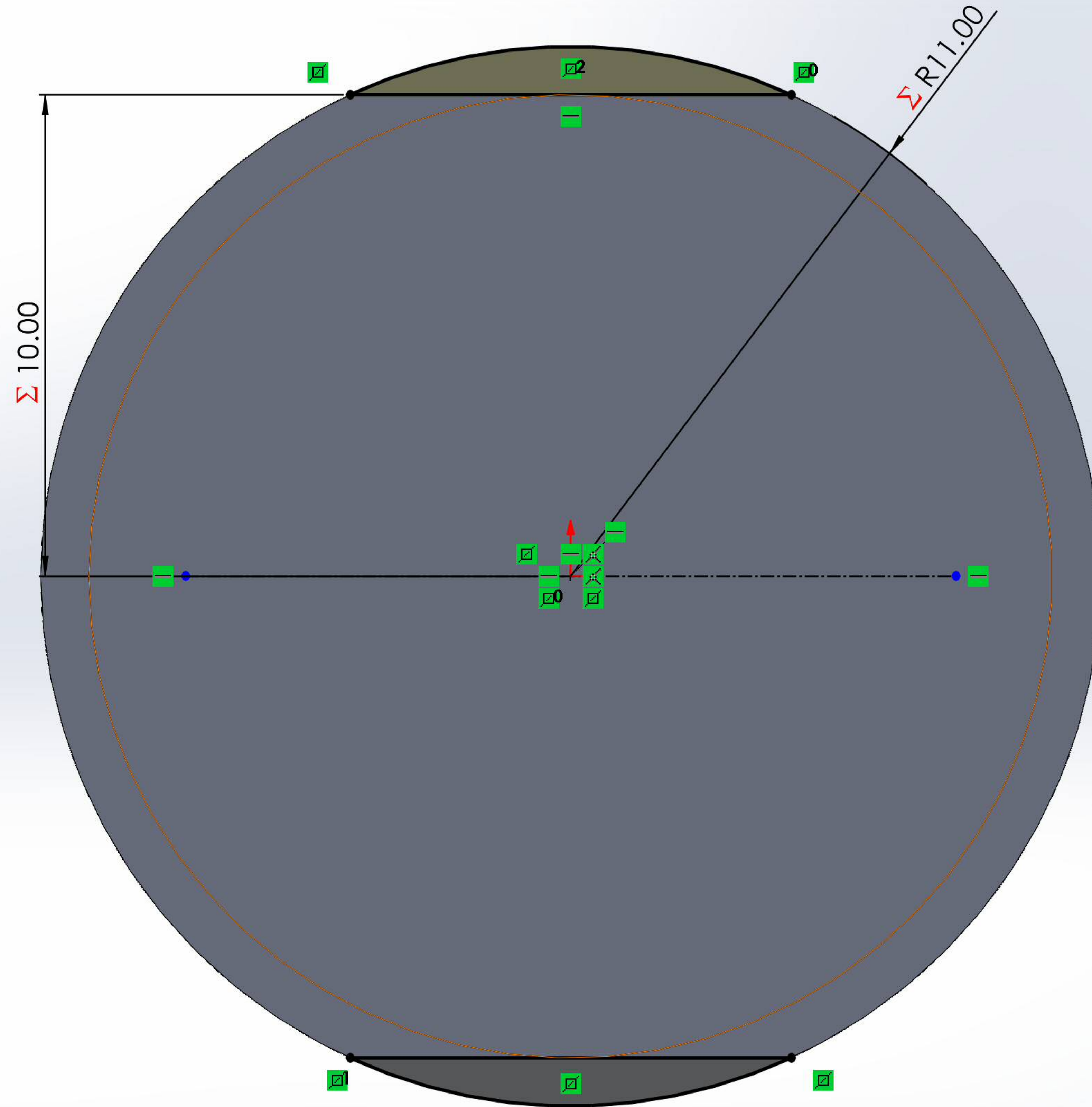
Sketch 2 (S&Th):



Chaitanya - Assignment 7.2 (Short and Thick)

- Solid Bodies(1)
 - Cut-Revolve1
- Equations
 - "Shaft Length" =45.00mm
 - "Brace Diameter" =22.00mm
 - "Shaft Diameter" =20.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 3 (S&Th):

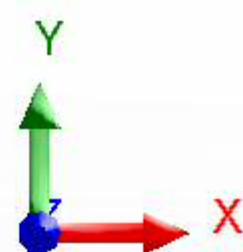
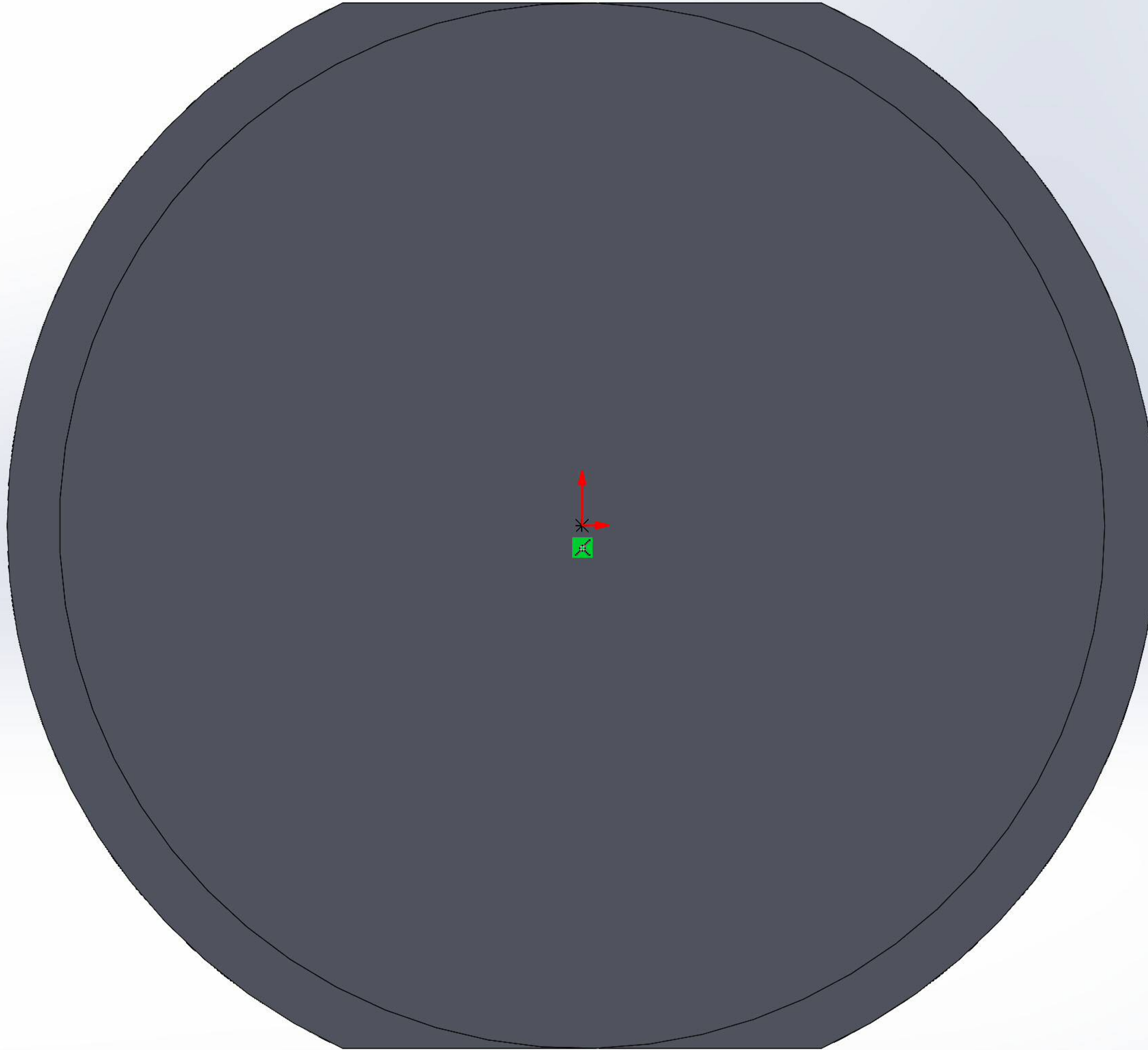


Navigation icons: Home, Table, Hide, Rotate, Color, Hide, Hide, Hide

Chaitanya - Assignment 7.2 (Short and Thick)

- Solid Bodies(1)
 - Cut-Extrude1
- Equations
 - "Shaft Length" =45.00mm
 - "Brace Diameter" =22.00mm
 - "Shaft Diameter" =20.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 4 (S&Th): Created By Hole Wizard



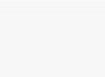
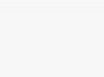
*Front

Chaitanya - Assignment 7.2 (Short and Thick)

- Solid Bodies(1)
 - Cut-Extrude1
- Equations
 - "Shaft Length" =45.00mm
 - "Brace Diameter" =22.00mm
 - "Shaft Diameter" =20.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5**
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 5 (S&Th): Created By Hole Wizard





Chaitanya - Assignment 7.2 (Short and Thick)

 Solid Bodies(1)

 Ø6.35mm Dowel Hole1

 Equations

 "Shaft Length" =45.00mm

 "Brace Diameter" =22.00mm

 "Shaft Diameter" =20.00mm

 Material <not specified>

 Front Plane

 Top Plane

 Right Plane

 Origin

 Revolve1

 Sketch1

 Cut-Revolve1

 Sketch2

 Cut-Extrude1

 Sketch3

 Ø6.35mm Dowel Hole1

 Sketch4

 Sketch5

 3/8-24 Tapped Hole1

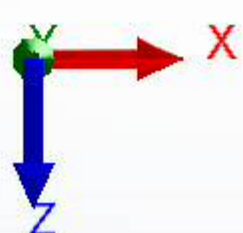
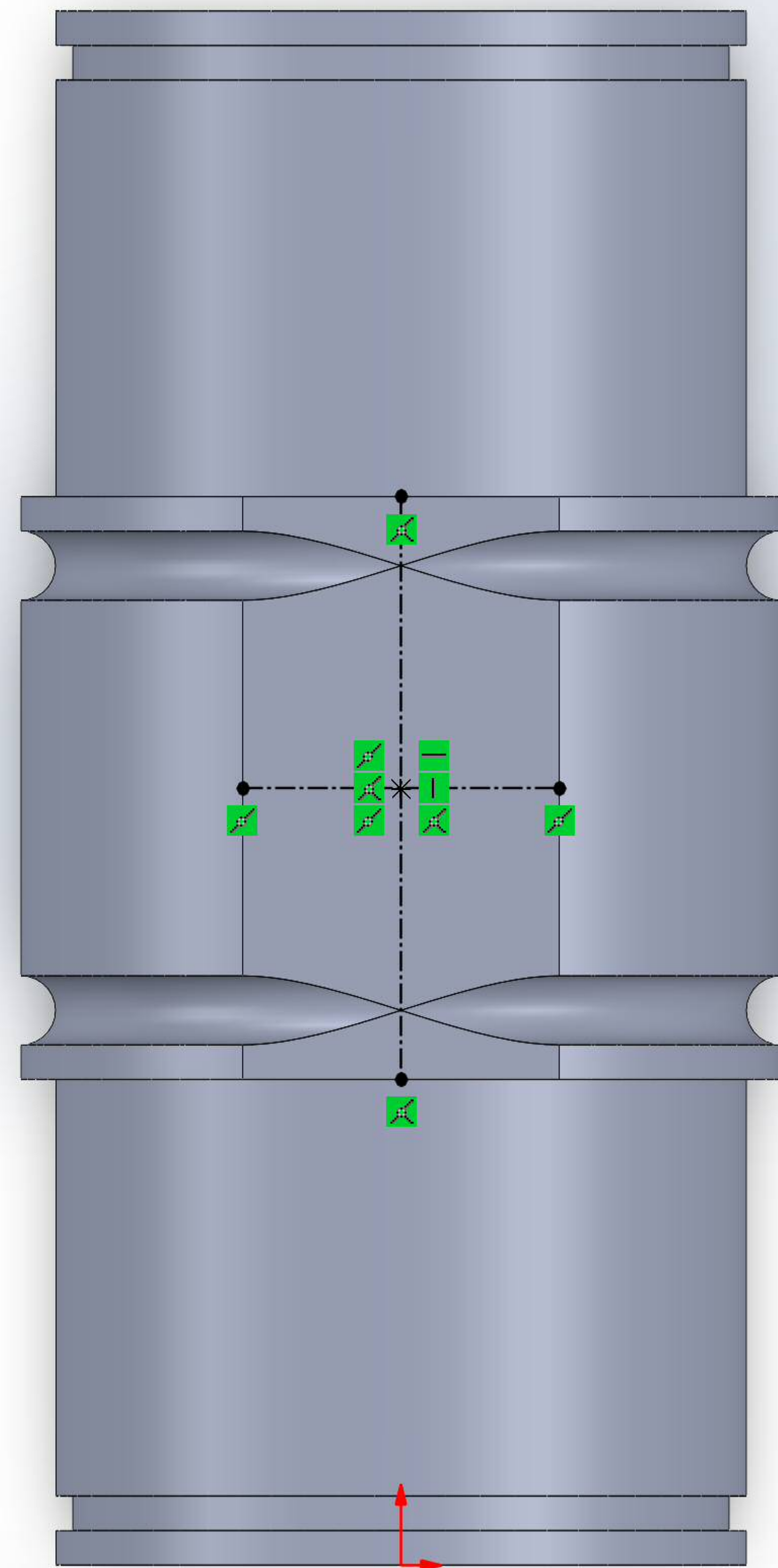
 Sketch6

 Sketch7

 Hole Thread1

 Chamfer1

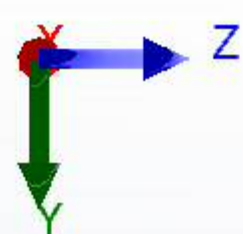
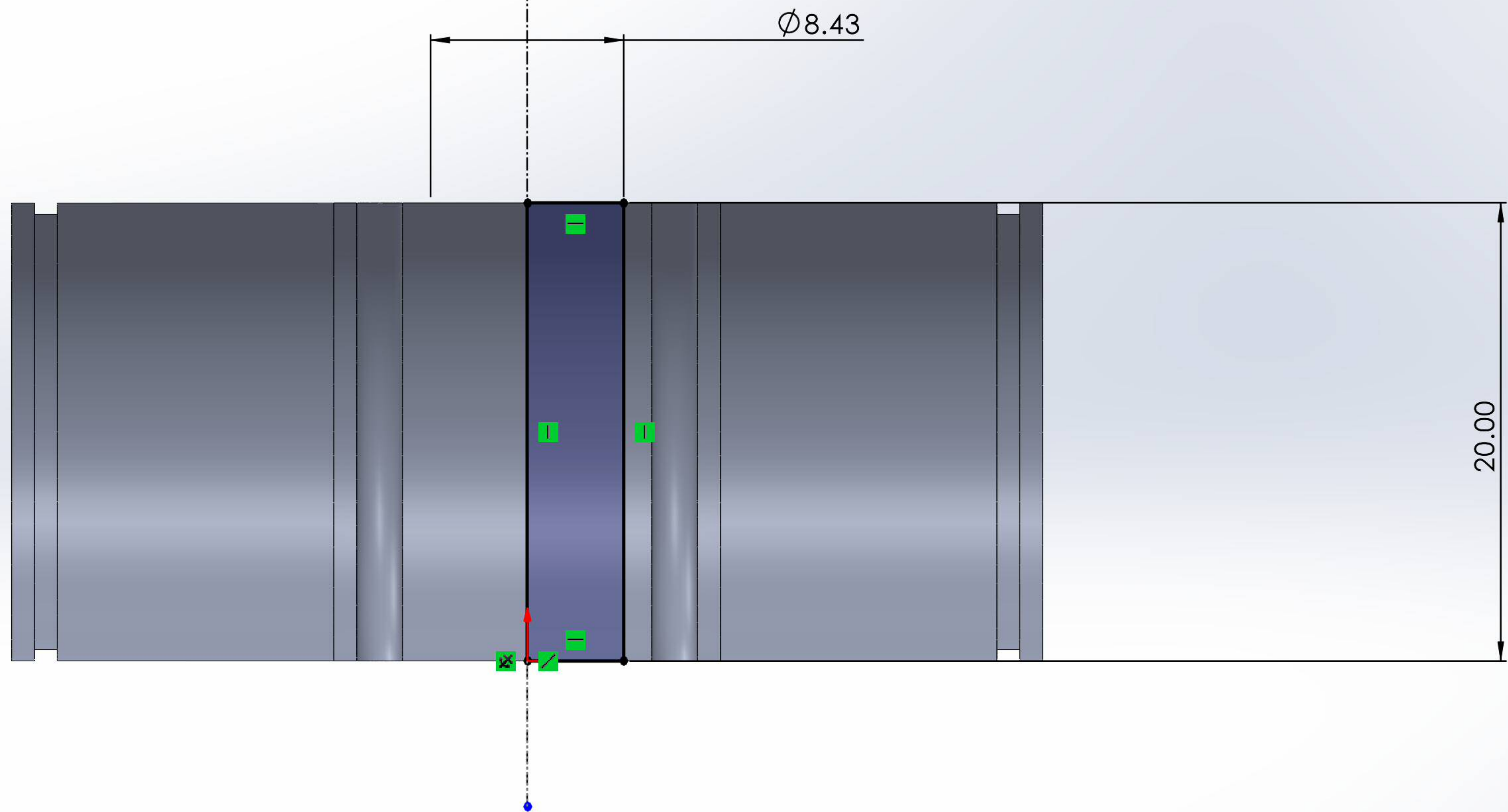
Sketch 6 (S&Th): Created For Hole Wizard



Chaitanya - Assignment 7.2 (Short and Thick)

- Solid Bodies(1)
 - Ø6.35mm Dowel Hole1
- Equations
 - "Shaft Length" =45.00mm
 - "Brace Diameter" =22.00mm
 - "Shaft Diameter" =20.00mm
- Material <not specified>
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Revolve1
 - Sketch2
- Cut-Extrude1
 - Sketch3
- Ø6.35mm Dowel Hole1
 - Sketch4
 - Sketch5
- 3/8-24 Tapped Hole1
 - Sketch6
 - Sketch7
 - Hole Thread1
- Chamfer1

Sketch 7 (S&Th): Created By Hole Wizard

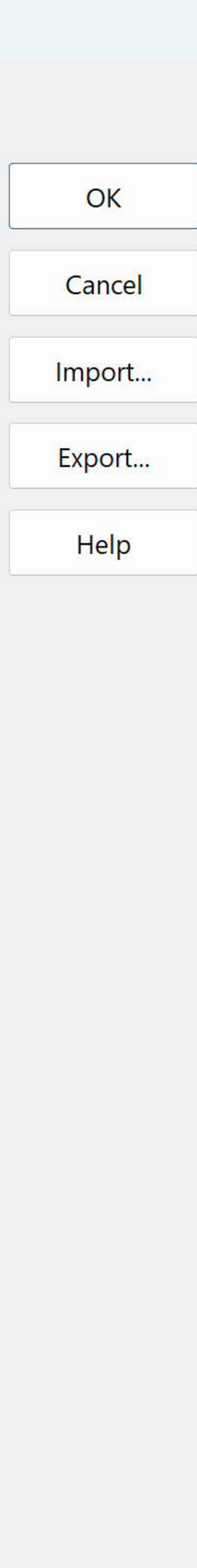
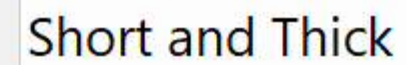


Global Variables & Equations (S&Th):

Variable / Equation	Evaluates to
Shaft Length	45.00mm
Shaft Diameter	22.00mm
Shaft Diameter	20.00mm
Shaft Length	45mm
Shaft Length * (3 / 8)	16.88mm
Shaft Diameter / 2	10mm
Shaft Diameter / 2	11mm
Shaft Diameter - "Shaft Diameter"	2mm
Shaft Diameter / 2	10mm
Shaft Diameter / 2	11mm
Shaft Diameter - "Shaft Diameter"	2mm

Degrees

☒ Automatic solve order

[illegible]

- OK
- Cancel
- Import...
- Export...
- Help

- OK
- Cancel
- Import...
- Export...
- Help

- OK
- Cancel
- Import...
- Export...
- Help

- OK
- Cancel
- Import...
- Export...
- Help

- OK
- Cancel
- Import...
- Export...
- Help

Equations, Global Variables, and Dimensions

Filter All Fields

Short and Thick

Name	Value / Equation	Evaluates to	Comments
Global Variables			
"Shaft Length"	= 45mm	45.00mm	
"Brace Diameter"	= 22mm	22.00mm	
"Shaft Diameter"	= 20mm	20.00mm	
Add global variable			
Features			
Add feature suppression			
Equations			
"D1@Sketch1"	= "Shaft Length"	45mm	
"D2@Sketch1"	= ("Shaft Length") * (3 / 8)	16.88mm	
"D3@Sketch1"	= "Shaft Diameter" / 2	10mm	
"D4@Sketch1"	= "Brace Diameter" / 2	11mm	
"D5@Sketch2"	= "Brace Diameter" - "Shaft Diameter"	2mm	
"D2@Sketch3"	= "Shaft Diameter" / 2	10mm	
"D1@Sketch3"	= "Brace Diameter" / 2	11mm	
"D4@Sketch2"	= "Brace Diameter" - "Shaft Diameter"	2mm	
Add equation			

Angular equation units: Degrees

Automatic solve order

Link to external file:

Equations, Global Variables, and Dimensions

Filter All Fields

Global Variables & Equations

Name	Value / Equation
Global Variables	
"Shaft Length"	= 45mm
"Brace Diameter"	= 22mm
"Shaft Diameter"	= 20mm
Add global variable	
Features	
Add feature suppression	
Equations	
"D1@Sketch1"	= "Shaft Length"
"D2@Sketch1"	= ("Shaft Length") * (3 / 8)
"D3@Sketch1"	= "Shaft Diameter" / 2
"D4@Sketch1"	= "Brace Diameter" / 2
"D5@Sketch2"	= "Brace Diameter" - "Shaft Diameter"
"D2@Sketch3"	= "Shaft Diameter" / 2
"D1@Sketch3"	= "Brace Diameter" / 2
"D4@Sketch2"	= "Brace Diameter" - "Shaft Diameter"
Add equation	

Angular equation units:

Degrees

Automatic solve order

Automatically rebuild

Link to external file:

Equations, Global Variables

Σ

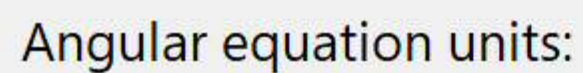




12

Name
 Global Variables
"Shaft Length"
"Brace Diameter"
"Shaft Diameter"
<i>Add global variable</i>
 Features
<i>Add feature sum</i>
 Equations
"D1@Sketch1"
"D2@Sketch1"
"D3@Sketch1"
"D4@Sketch1"
"D5@Sketch2"
"D2@Sketch3"
"D1@Sketch3"
"D4@Sketch2"
<i>Add equation</i>

☐ Automatically rebuild
 ☐ Link to external file:

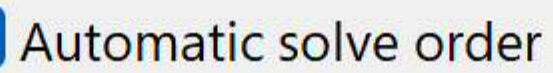


Global

e / Equation

mm
mm
mm
Shaft Length"
Shaft Length"
Shaft Diameter
Face Diameter
Face Diameter
Shaft Diameter
Face Diameter
Face Diameter

Degrees



Equations, Global Variables, and Dimensions

Filter All Fields

Global Variables & Equations

Name	Value / Equation
Global Variables	
"Shaft Length"	= 45mm
"Brace Diameter"	= 22mm
"Shaft Diameter"	= 20mm
Add global variable	
Features	
Add feature suppression	
Equations	
"D1@Sketch1"	= "Shaft Length"
"D2@Sketch1"	= ("Shaft Length") * (3 / 8)
"D3@Sketch1"	= "Shaft Diameter" / 2
"D4@Sketch1"	= "Brace Diameter" / 2
"D5@Sketch2"	= "Brace Diameter" - "Shaft Diameter"
"D2@Sketch3"	= "Shaft Diameter" / 2
"D1@Sketch3"	= "Brace Diameter" / 2
"D4@Sketch2"	= "Brace Diameter" - "Shaft Diameter"
Add equation	

Angular equation units:

Degrees

Automatic solve order

Automatically rebuild

Link to external file:

